

Baseline Hard Drug Use and Two-Year HAART Response in HIV-Infected Men

Guangyu Qiu

Introduction

Population: 463/715 HIV-infected men initiating HAART

Objective: Assess whether 2-year outcomes improve and whether baseline hard drug use modifies these changes.

Outcomes (continuous, change from baseline to 2 years):

- $\log_{10}$ Viral load
- $CD4^+$ T cell count
- Physical health score
- Mental health score

Exposure: Baseline hard drug use (yes/no)

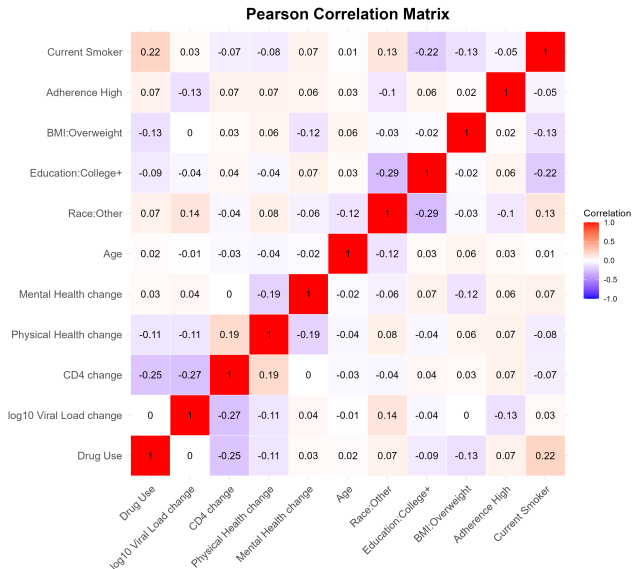
Confounders: Age, race/ethnicity, education, BMI, smoking

Sensitivity check: Adherence

Baseline Characteristics by Hard Drug Use (Selected Variables)

Characteristic	No (N = 427)	Yes (N = 36)	P-value
BMI category			0.019
Normal	205 (48%)	26 (72%)	
Overweight	208 (49%)	9 (25%)	
Underweight	14 (3.3%)	1 (2.8%)	
Smoking			< 0.001
Current smoker	149 (35%)	27 (75%)	
Non-current	278 (65%)	9 (25%)	
Change in log ₁₀ Viral Load	-2.74 (1.22)	-2.72 (1.32)	> 0.9
Change in CD4 count	181.64 (175.10)	11.18 (203.64)	< 0.001
Change in Physical Health	-1.34 (8.17)	-4.80 (8.38)	0.022
Change in Mental Health	2.42 (11.75)	3.89 (15.65)	0.6

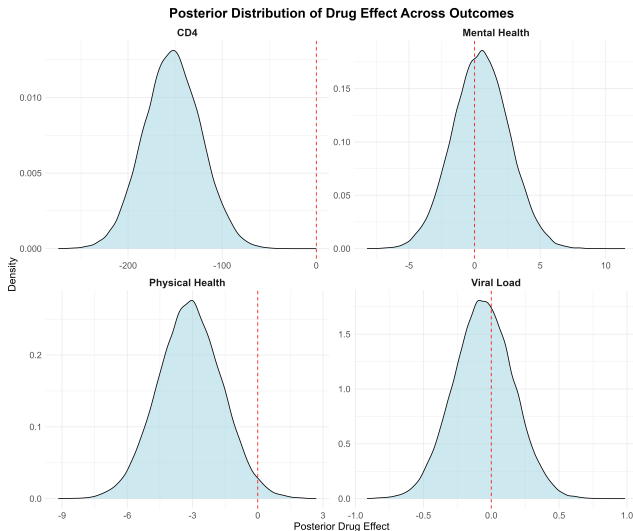
Correlation Matrix of Outcomes and Covariates



Bayesian Model Results

Outcome	Post. Est.	95% PI	P-value	WAIC
Δ VLOAD	-0.057	(-0.486, 0.375)	0.994	1513
Δ VLOAD+adh	-0.005	(-0.437, 0.427)	0.989	1512
Δ CD4	-152.8	(-211.6, -92.8)	< 0.001	6126
Δ CD4+adh	-157.0	(-216.4, -97.1)	< 0.001	6120
Δ PHYS	-3.12	(-5.99, -0.24)	0.222	3270
Δ PHYS+adh	-3.31	(-6.16, -0.40)	0.182	3264
Δ MENT	0.48	(-3.74, 4.72)	0.875	3628
Δ MENT+adh	0.18	(-4.13, 4.48)	0.840	3631

Posterior Density of Main Effects



Study Limitations

- Missing data were handled via complete-case analysis, which may introduce bias if data are missing not at random (MNAR).
- Baseline hard drug users represented only 7.7% of the sample, resulting in a small effective sample size for the exposed group; this may limit statistical power in t-tests, chi-squared tests, and regression analyses.
- Most control variables included in the current models showed little or no significant association with the outcomes; future work could explore additional potential confounders and apply formal variable selection to improve model robustness.